

Abin Anil

Python Backend Engineer | REST APIs, Async Services & Cloud Deployment

Kottayam, Kerala, India | +91-9656768702 | abinanil000@gmail.com
[linkedin.com/in/abinanil000](https://www.linkedin.com/in/abinanil000) | github.com/abinanil000datascience

Professional Summary

Python backend developer who has independently built asynchronous REST APIs (FastAPI, Django), resolved asyncio concurrency failures, and deployed containerized services to GCP Cloud Run. Backed by 2+ years of enterprise Technical Support Engineering – API debugging, SQL, observability (Kibana, Grafana), incident response, and root-cause analysis on a live production platform sustaining 99.8% uptime. Experienced with PostgreSQL/pgvector data layers, Docker, and Jenkins CI/CD.

Technical Skills

Programming: Python, SQL, JavaScript, OOP, Data Structures & Algorithms

Backend Development: FastAPI, Django, Flask, REST API Design, Third-Party API Integration (OpenAI API), Asynchronous Programming (asyncio), Concurrency & Multi-Threading

Databases: PostgreSQL, SQL & NoSQL, pgvector, SQLAlchemy, pycopg, Hybrid Search (BM25 + Vector Indexing/HNSW)

Cloud/DevOps: Docker, Docker Compose, GCP Cloud Run, Jenkins CI/CD, Git, GitHub, AI-Assisted Development (GitHub Copilot: test generation, refactoring, code review assistance)

Reliability/Observability: API Debugging, Root Cause Analysis, Incident Management, Kibana, Grafana, Postman, Swagger, Network Traffic Inspection

Frontend Integration (working knowledge): React, Vite, Tailwind CSS, Recharts

Applied ML/AI (domain exposure): Python-based ML pipelines, Hugging Face, LangChain/LangGraph service integration, Vector Embeddings, Retrieval-Augmented Generation (RAG), OpenAI API Integration, Generative AI, Generative AI Agents, Large Language Modeling, LLM Application Development, Tool Calling, Prompt Engineering, Prompt Patterns, Context Engineering – used as backend building blocks, not the core specialization

Projects

Multi-Agent Service Orchestration Platform (Nexus Intelligence)

Sep 2026

GitHub: <https://github.com/abinanil000datascience/nexus-ai-trading-engine>

- Rebuilt a single-threaded Streamlit prototype into a fully decoupled, asynchronous backend on FastAPI, exposing four independent service workers (Scout, Quant, Analyst, Gatekeeper) behind a unified API layer.
- Diagnosed and eliminated critical asyncio event-loop collisions between FastAPI and background execution threads, restructuring endpoints to isolate long-running workloads without blocking the request/response cycle.
- Integrated a real-time data ingestion layer (streaming market data via Pandas) with downstream API consumers, processing data continuously alongside incoming API requests.
- Built and served a REST API layer for a React/Vite/Tailwind CSS dashboard, delivering live service telemetry and computed results to the frontend in real time.

Tech Stack: Python, FastAPI, asyncio, Pandas, REST APIs, React, Vite, Tailwind CSS, Recharts.

IT Ticket Resolution Backend (Healthcare ITSM Engine)

Aug 2026

GitHub: https://github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Designed a sub-linear hybrid search architecture in PostgreSQL/pgvector, combining keyword matching (tsvector/BM25) with HNSW-indexed vector search to accelerate retrieval across a historical ticket corpus.
- Built and deployed an asynchronous FastAPI backend serving retrieval and classification results, fully containerized with Docker for reproducible deployment.
- Eliminated ongoing third-party API costs by replacing a cloud-dependent inference call with a self-hosted local model service, cutting external dependencies from the request path.

Tech Stack: PostgreSQL, pgvector, HNSW, FastAPI, Docker, Python, Hugging Face.

Applicant Matching Platform Backend (Enterprise AI Career Platform)

Aug 2026

GitHub: <https://github.com/abinanil000datascience/ai-career-platform>

- Built an asynchronous FastAPI backend with a PostgreSQL/pgvector data layer to serve resume-matching and applicant-tracking workflows.
- Designed the API and data schema for storing, querying, and ranking structured applicant data.
- Deployed the containerized service to GCP Cloud Run, with Docker-based build and release packaging.

Tech Stack: Python, FastAPI, PostgreSQL, pgvector, Docker, GCP Cloud Run.

- Built a Django REST API backend serving real-time delay-risk predictions to a React front end, on a 97% on-time, class-imbalanced dataset.
- Designed the API contract and response schema to give end users real-time visibility into at-risk shipments.

Tech Stack: Python, Django, REST APIs, React, Scikit-Learn.

Professional Experience

Technical Support Engineer, Feathersoft

Aug 2024 – Present

Distributed DNA diagnostics platform (DDC)

- **API & Database Operations:** Managed production reliability for a distributed DNA diagnostics platform (DDC), ensuring data integrity across complex probability (PI) calculations and automated report generation workflows, sustaining 99.8% system uptime.
- Reduced production error rates by 22% by debugging API-layer failures and data-ingestion bottlenecks through network traffic inspection, cURL requests, and payload validation using Postman and Swagger.
- Spearheaded system observability and anomaly detection by analyzing application logs in Kibana and Grafana to isolate root causes of complex workflow disruptions, accelerating incident resolution time by 15%.
- Reduced Mean Time to Resolution (MTTR) by 20% by executing direct SQL (SELECT/UPDATE) database interventions to resolve live data discrepancies and maintain the structural integrity of diagnostic reporting.
- Engineered and validated temporary diagnostic workflows within staging and test environments, acting as a technical bridge between end-users and the core development team to unblock production.
- Led cross-functional incident triage with US-based enterprise clients and engineering teams, translating complex diagnostic logs into actionable technical fixes, sustaining a 99% CSAT rating across 120+ escalated incidents.

Professional Training

Data Science With AI Professional Certification – MetaiLearn

Nov 2025 – Jun 2026

Structured part-time program, completed alongside full-time role

- **Backend & Data Foundations:** Completed Python for Data Science (OOP, data structures & algorithms, debugging), relational and NoSQL databases, and querying with SQL & NoSQL.
- **API & Full-Stack Development:** Studied Flask and Streamlit applications with Git-based deployment, and integrating the OpenAI API into backend applications (endpoints, response and error handling, token usage).
- **AI-Assisted Development:** Practiced GitHub Copilot workflows for unit test generation, debugging, refactoring, and code review assistance, plus RAG and LLM pipeline fundamentals.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Dec 2020 – May 2024

University College of Engineering, Thodupuzha

CGPA: 7.33/10.0

Certifications & Training

- **Fundamentals of AI Agents Using RAG and LangChain** – IBM Sep 2026
Credential ID: S0HTSWVNLSTW | [Verify Credential](#)
- **Python Programming & Machine Learning Intensive Program** – Reverttech IT Solutions (1 month) Dec 2024
- **Advanced Machine Learning Workshop** – Techmaghi Training Institute Sep 2022

Achievements

- **Employee of the Month (January 2026)** – Recognized for technical excellence, critical root-cause analysis, and outstanding client communication.